

Series with non-negative terms



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Comparison test

Introduction: Comparison test

- ▶ In this chapter we study **series with non-negative terms**, for which we have very effective tools to determine their **behavior**
- ▶ The first tool is the **comparison test**: we compare the series under study with a series whose behavior we already know
- ▶ As a first application, we will prove that the series $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ diverges for $\alpha \leq 1$

Series with non-negative terms

- ▶ A series $\sum a_k$ **with non-negative terms** (or positive terms) is **regular**, that is, it is either convergent or divergent to $+\infty$ (it cannot be irregular, as we have proved)
- ▶ Such a series converges if and only if the sequence of its partial sums is bounded

Comparison test

Theorem 1: Comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with non-negative terms such that $a_k \leq b_k$ eventually; then:

$$i) \quad \sum b_k \text{ convergent} \Rightarrow \sum a_k \text{ convergent}$$

$$ii) \quad \sum a_k \text{ divergent} \Rightarrow \sum b_k \text{ divergent}$$

- ▶ The series $\sum b_k$ is called the **dominating series** (majorant), while the series $\sum a_k$ is called the **dominated series** (minorant).

Proof (1/2)

Since $a_k \leq b_k$ eventually, there exists $m \in \mathbb{N}$ such that:

$$a_k \leq b_k, \quad \forall k \geq m$$

Let us now consider the n -th partial sums of the tails of the two sequences $\{a_k\}$ and $\{b_k\}$, with $n > m$:

$$s_n^a = \sum_{k=m+1}^n a_k \quad \text{and} \quad s_n^b = \sum_{k=m+1}^n b_k$$

Since $0 \leq a_k \leq b_k$, $\forall k \geq m$, we have

$$0 \leq s_n^a \leq s_n^b, \quad \forall n \geq m \quad (*)$$

Proof (2/2)

- ▶ The sequences $\{s_n^a\}$ and $\{s_n^b\}$ are regular, since $\{a_k\}$ and $\{b_k\}$ have non-negative terms. Hence the claims *i*) and *ii*) are logically equivalent, so it suffices to prove the second one.
- ▶ Saying that $\sum a_n$ is divergent means, by the definition of divergent series, that $s_n \rightarrow +\infty$ as $n \rightarrow +\infty$. Moreover, since every tail has the same behavior as the original series, we have $s_n^a \rightarrow +\infty$ as $n \rightarrow +\infty$.
- ▶ From $(*)$, by the comparison theorem for sequences, also $s_n^b \rightarrow +\infty$ as $n \rightarrow +\infty$. Hence the tail of $\{b_k\}$ is divergent and consequently $\sum b_n$ is divergent.



The series $\sum 1/k^\alpha$ with $\alpha \leq 1$

Observation 1

Given $\alpha \leq 1$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent to $+\infty$.

Proof

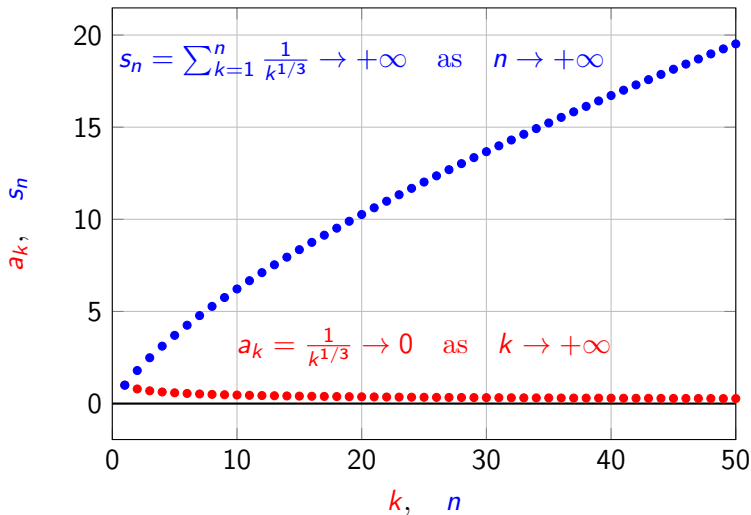
With $\alpha = 1$ we have the harmonic series, which diverges to $+\infty$.

With $\alpha < 1$, the series dominates the harmonic series, since:

$$\frac{1}{k} \leq \frac{1}{k^\alpha}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha < 1$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is divergent by the comparison test. □

Graphical interpretation: $\alpha = \frac{1}{3}$



Summary

Series with non-negative terms:

- ▶ They are always regular: they converge or diverge to $+\infty$
- ▶ They converge if and only if the sequence of partial sums is bounded

Comparison test:

- ▶ If $0 \leq a_k \leq b_k$ eventually: convergent dominating series $\Rightarrow$ convergent dominated series; divergent dominated series $\Rightarrow$ divergent dominating series
- ▶ Consequence: $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ diverges to $+\infty$ for $\alpha \leq 1$

Limit comparison test



Introduction: Limit comparison test

- ▶ It is often not easy to find an inequality $a_k \leq b_k$, while it is simple to find a sequence **asymptotically equivalent** to the general term
- ▶ The **limit comparison test** states that two series with positive terms and asymptotically equivalent general terms have the **same behavior**
- ▶ We will use it to prove that $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ converges for $\alpha \geq 2$ and to study a first example

Theorem 2: Limit comparison test

Let $\{a_k\}$ and $\{b_k\}$ be two sequences with positive terms. If the sequences are asymptotically equivalent, that is, if:

$$a_k \sim b_k \quad \text{as } k \rightarrow +\infty$$

then the corresponding series $\sum a_k$ and $\sum b_k$ are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

Proof (1/2)

The series $\sum a_k$ and $\sum b_k$ are regular because the sequences $\{a_k\}$ and $\{b_k\}$ have positive terms. Since $a_k \sim b_k$ as $k \rightarrow +\infty$, we have:

$$\frac{a_k}{b_k} \rightarrow 1 \quad \text{as } k \rightarrow +\infty$$

Hence, for every $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$ we have:

$$1 - \varepsilon < \frac{a_k}{b_k} < 1 + \varepsilon \quad \text{and hence} \quad (1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$$

since $b_k > 0$, $\forall k$.

Proof (2/2)

- ▶ We have thus proved that $(1 - \varepsilon) b_k < a_k < (1 + \varepsilon) b_k$ eventually.
- ▶ Hence, by the comparison test, the series $\sum a_k$ and $\sum b_k$ have the same behavior.
- ▶ The first of the two inequalities implies that if $\sum a_k$ is convergent then $\sum b_k$ is also convergent, while the second one implies that if $\sum a_k$ is divergent then $\sum b_k$ is also divergent.



The series $\sum 1/k^\alpha$ with $\alpha \geq 2$

Observation 2

Given $\alpha \geq 2$, the series $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent.

Proof

With $\alpha = 2$ we have:

$$\frac{1}{k^2} \sim \frac{1}{k(k+1)} \quad \text{as } k \rightarrow +\infty$$

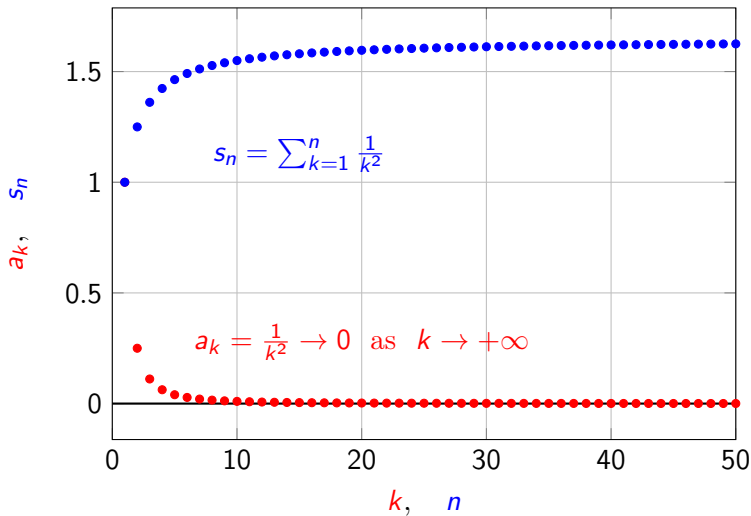
and the series $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$ converges (Mengoli's series). Hence, by the limit comparison test, $\sum_{k=1}^{\infty} \frac{1}{k^2}$ also converges.

With $\alpha > 2$, the series is dominated by the series $\sum_{k=1}^{\infty} \frac{1}{k^2}$, since:

$$\frac{1}{k^\alpha} \leq \frac{1}{k^2}, \quad \forall k \in \mathbb{N}, k \geq 1 \quad \text{and} \quad \alpha > 2$$

hence $\sum_{k=1}^{\infty} \frac{1}{k^\alpha}$ is convergent by the comparison test. □

Graphical interpretation: $\alpha = 2$



Example (1/2)

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{5k + \cos k}{3 + 2k^3}$$

We have:

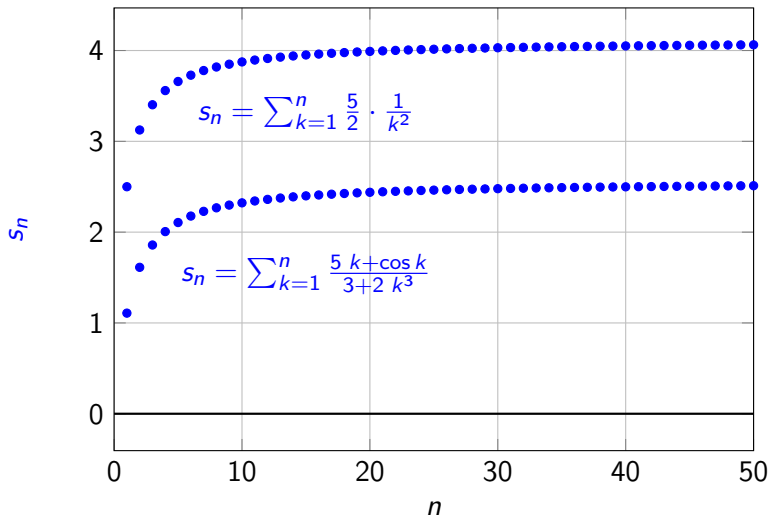
$$\frac{5k + \cos k}{3 + 2k^3} \sim \frac{5}{2} \cdot \frac{1}{k^2} \quad \text{as } k \rightarrow +\infty$$

and

$$\sum_{k=1}^{\infty} \frac{5}{2} \cdot \frac{1}{k^2} \quad \text{converges}$$

hence the series is convergent by the limit comparison test.

Example (2/2)



Warning

However, even if the sequences are asymptotically equivalent, the corresponding series may not have the same sum.

- ▶ In the previous example, both series converge but their sums are different.

Summary

Limit comparison test:

- ▶ If $a_k, b_k > 0$ and $a_k \sim b_k$ as $k \rightarrow +\infty$, then $\sum a_k$ and $\sum b_k$ have the same behavior
- ▶ The sums of the two series may however be different

Consequences:

- ▶ $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ converges for $\alpha \geq 2$ (comparison with Mengoli's series)
- ▶ To study a series it suffices to find a simpler asymptotically equivalent term

Condensation test

Introduction: Condensation test

- ▶ So far we know that $\sum \frac{1}{k^\alpha}$ diverges for $\alpha \leq 1$ and converges for $\alpha \geq 2$: the case $1 < \alpha < 2$ remains open
- ▶ The **condensation test** replaces the series $\sum a_k$ with the “condensed” series $\sum 2^k a_{2^k}$, which only uses the terms with index 2^k
- ▶ For decreasing terms the two series have the same behavior: the proof relies on two inequalities between partial sums, proved by induction

Theorem 3: Condensation test

If $\{a_k\}$ is an eventually decreasing sequence with non-negative terms, then the series

$$\sum_{k=1}^{\infty} a_k \quad \text{and} \quad \sum_{k=0}^{\infty} 2^k a_{2^k}$$

are regular and have the same behavior, i.e., either they are both convergent or they are both divergent.

Partial sums $\tilde{s}_n$ and s_n

- ▶ Before proving the theorem, let us consider the sequence $\{\tilde{s}_n\}$ of the partial sums of the sequence $\{2^k a_{2^k}\}$, that is, the sequence:

$$\tilde{s}_n = \sum_{k=0}^n 2^k a_{2^k} \quad \forall n \geq 0$$

- ▶ Let us derive two important relations with the sequence $\{s_n\}$ of the partial sums of the sequence $\{a_k\}$, that is, the sequence:

$$s_n = \sum_{k=1}^n a_k \quad \forall n \geq 1$$

Partial sums s_{2^n-1}

- We observe that, for certain values of n , the partial sums s_n are:

$$\underbrace{s_1}_{= s_{2^1-1}} = a_1$$

$$\underbrace{s_3}_{= s_{2^2-1}} = s_1 + a_2 + \underbrace{a_3}_{\leq a_2} \leq a_1 + 2 a_2$$

$$\underbrace{s_7}_{= s_{2^3-1}} = s_3 + a_4 + \underbrace{a_5}_{\leq a_4} + \underbrace{a_6}_{\leq a_4} + \underbrace{a_7}_{\leq a_4} \leq a_1 + 2 a_2 + 4 a_4$$

$$\underbrace{s_{15}}_{= s_{2^4-1}} = s_7 + a_8 + \underbrace{a_9}_{\leq a_8} + \cdots + \underbrace{a_{15}}_{\leq a_8} \leq a_1 + 2 a_2 + 4 a_4 + 8 a_8$$

First inequality

Observation 3

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$s_{2^n-1} \leq \underbrace{\sum_{k=0}^{n-1} 2^k a_{2^k}}_{=\tilde{s}_{n-1}} \quad \forall n \geq 1 \quad (\text{P1})$$

Proof (1/2)

We prove by induction on 2^n that

$$s_{2^n-1} \leq \sum_{k=0}^{n-1} 2^k a_{2^k} \quad \forall n \geq 1$$

Base case. Let $n = 1$. Then the claim becomes

$$s_{2^1-1} \leq 2^0 a_1$$

i.e., $a_1 \leq a_1$, which is clearly true.

Proof (2/2)

Inductive step. Assume that it is true for 2^{n-1} and let us prove it for 2^n . By the inductive hypothesis, we have

$$s_{2^{n-1}-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k}$$

Moreover, we have:

$$\begin{aligned} s_{2^n-1} &= s_{2^{n-1}-1} + \underbrace{\sum_{i=2^{n-1}}^{2^n-1} a_i}_{\leq 2^{n-1} a_{2^{n-1}}} \\ &\leq 2^{n-1} a_{2^{n-1}} \end{aligned}$$

hence

$$s_{2^n-1} \leq \sum_{k=0}^{n-2} 2^k a_{2^k} + 2^{n-1} a_{2^{n-1}} = \sum_{k=0}^{n-1} 2^k a_{2^k}$$

which is exactly the desired claim for 2^n .



Partial sums $\tilde{s}_n$

- We now observe that the values of $\tilde{s}_n$ are:

$$\tilde{s}_0 = a_1 \leq 2 a_1 = 2 \underbrace{s_1}_{= s_{2^0}}$$

$$\tilde{s}_1 = \tilde{s}_0 + 2 a_2 \leq 2 s_1 + 2 a_2 = 2 \underbrace{s_2}_{= s_{2^1}}$$

$$\tilde{s}_2 = \tilde{s}_1 + 4 a_4 \leq 2 s_2 + 2 a_3 + 2 a_4 = 2 \underbrace{s_4}_{= s_{2^2}}$$

(since $a_4 \leq a_3$)

$$\tilde{s}_3 = \tilde{s}_2 + 8 a_8 \leq 2 s_4 + 2 a_5 + \cdots + 2 a_8 = 2 \underbrace{s_8}_{= s_{2^3}}$$

(since $a_8 \leq a_7 \leq a_6 \leq a_5$)

Second inequality

Observation 4

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have:

$$\underbrace{\sum_{k=0}^n 2^k a_{2^k}}_{=\tilde{s}_n} \leq 2 s_{2^n} \quad \forall n \geq 0 \quad (\text{P2})$$

Proof (1/2)

We prove by induction on 2^n that

$$\tilde{s}_n \leq 2 s_{2^n} \quad \forall n \geq 0$$

Base case. Let $n = 0$. Then the claim becomes

$$\tilde{s}_0 \leq 2 s_{2^0} = 2 s_1$$

i.e., $a_1 \leq 2 a_1$, which is clearly true.

Proof (2/2)

Inductive step. Assume that it is true for $n - 1$ and let us prove it for n . By the inductive hypothesis, we have

$$\tilde{s}_{n-1} \leq 2 s_{2^{n-1}}$$

Moreover, we have:

$$\begin{aligned} \tilde{s}_n &= \tilde{s}_{n-1} + \underbrace{2^n a_{2^n}}_{2^n} \\ &\leq \sum_{i=2^{n-1}+1}^{2^n} 2 a_i \end{aligned}$$

hence

$$\tilde{s}_n \leq 2 s_{2^{n-1}} + \sum_{i=2^{n-1}+1}^{2^n} 2 a_i = 2 s_{2^n}$$

which is exactly the desired claim for n . □

We are now able to prove the condensation test.

Proof (1/2)

From relation (P1) we have:

$$s_{2^n-1} \leq \tilde{s}_{n-1}, \quad \forall n \geq 1$$

- ▶ Hence, if $\{\tilde{s}_n\}$ is convergent, that is, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, the subsequence $\{s_{2^n-1}\}$ is bounded.
- ▶ Since $\{s_n\}$ is monotone, the whole sequence $\{s_n\}$ is bounded and convergent by the monotone sequence theorem.
- ▶ Consequently, if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is convergent, then $\sum_{k=1}^{\infty} a_k$ is also convergent.

Proof (2/2)

From relation (P2) we have:

$$\tilde{s}_n \leq 2 s_{2^n}, \quad \forall n \geq 0$$

- ▶ Hence, if $\{s_n\}$ is convergent, that is, if $\sum_{k=1}^{\infty} a_k$ is convergent, then the subsequence $\{s_{2^n}\}$ is convergent (since $\{s_n\}$ is monotone). Consequently, by comparison, $\{\tilde{s}_n\}$ is also convergent, that is, $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.
- ▶ Moreover, since $\{a_k\}$ has non-negative terms, the series cannot be irregular; consequently, we also have:

$$\sum_{k=1}^{\infty} a_k \text{ divergent} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ divergent}$$

which completes the proof of the theorem.



Necessary and sufficient condition

Given an eventually decreasing sequence $\{a_k\}$ with non-negative terms, we have proved that:

$$\sum_{k=1}^{\infty} a_k \text{ conv./div.} \iff \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ conv./div.}$$

- Hence the convergence/divergence of $\sum_{k=0}^{\infty} 2^k a_{2^k}$ is a necessary and sufficient condition for the convergence/divergence of $\sum_{k=1}^{\infty} a_k$.

Summary

Condensation test:

- ▶ If $\{a_k\}$ is eventually decreasing with non-negative terms, $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=0}^{\infty} 2^k a_{2^k}$ have the same behavior

Idea of the proof:

- ▶ (P1): $s_{2^n-1} \leq \tilde{s}_{n-1}$ and (P2): $\tilde{s}_n \leq 2 s_{2^n}$, proved by induction
- ▶ The partial sums are monotone: we conclude with the monotone sequence theorem and by comparison

Generalized harmonic series

Introduction: Generalized harmonic series

- ▶ The series $\sum \frac{1}{k^\alpha}$ are the reference series most frequently used in comparisons
- ▶ Thanks to the condensation test we can now determine their behavior for **every** $\alpha \in \mathbb{R}$
- ▶ The result reduces to the study of a **geometric series**

Generalized harmonic series

Definition 1: Generalized harmonic series

Given $\alpha \in \mathbb{R}$, the series

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$$

is called the *generalized harmonic series*

Behavior of the generalized harmonic series

Theorem 4: Behavior of the generalized harmonic series

Given $\alpha \in \mathbb{R}$, the generalized harmonic series

$$\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}} \quad \text{is} \quad \begin{cases} \text{divergent to } +\infty & \text{if } \alpha \leq 1 \\ \text{convergent} & \text{if } \alpha > 1 \end{cases}$$

Proof

We have already established that it diverges for $\alpha \leq 1$.

For $\alpha > 1$, the sequence $\left\{ \frac{1}{k^\alpha} \right\}$ is a decreasing sequence with positive terms. We have:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^\alpha} = \sum_{k=0}^{\infty} (2^{1-\alpha})^k$$

that is, a geometric series with ratio $2^{1-\alpha}$, which converges if and only if $2^{1-\alpha} < 1$, that is, if $1 - \alpha < 0$.

Hence, for $\alpha > 1$, by the condensation test, the series converges.



Summary

Generalized harmonic series $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$:

- ▶ It diverges to $+\infty$ if $\alpha \leq 1$ (comparison test with the harmonic series)
- ▶ It converges if $\alpha > 1$ (condensation test: we obtain the geometric series with ratio $2^{1-\alpha}$)

Use:

- ▶ It is the reference series for the comparison and limit comparison tests

Passing from the discrete to the continuous



Introduction: Passing from the discrete to the continuous

- ▶ The tools used to establish asymptotic estimates of functions can also be used to obtain asymptotic estimates of sequences (**passing from the discrete to the continuous**)
- ▶ They provide useful tools for studying the behavior of a series with positive terms
- ▶ We see three examples in which asymptotic equivalences and Maclaurin expansions, combined with the limit comparison test, determine the behavior of the series

Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \frac{e^{1/k} - 1}{k}$$

From the asymptotic equivalence

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x) \quad \text{as} \quad \varepsilon(x) \rightarrow 0$$

we have

$$\frac{e^{1/k} - 1}{k} \sim \frac{1}{k^2} \quad \text{as} \quad k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/k^2$.

Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \log \left(\frac{k+3}{k+2} \right)$$

We have $\frac{k+3}{k+2} = 1 + \frac{1}{k+2}$. From the asymptotic equivalence

$$\log(1 + \varepsilon(x)) \sim \varepsilon(x) \quad \text{as} \quad \varepsilon(x) \rightarrow 0$$

we have

$$\log \left(1 + \frac{1}{k+2} \right) \sim \frac{1}{k+2} \sim \frac{1}{k} \quad \text{as} \quad k \rightarrow +\infty$$

Thus it is a series with positive terms whose general term is asymptotically equivalent to $1/k$. By comparison with the harmonic series, this series diverges to $+\infty$.

Example

Let us determine the behavior of the series:

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \sin \frac{1}{k} \right)$$

From the third-order Maclaurin expansion

$$\sin(\varepsilon(x)) = \varepsilon(x) - \frac{(\varepsilon(x))^3}{3!} + o((\varepsilon(x))^3) \quad \text{as } \varepsilon(x) \rightarrow 0$$

we have

$$\frac{1}{k} - \sin \frac{1}{k} = \frac{1}{k} - \left(\frac{1}{k} - \frac{1}{6k^3} + o\left(\frac{1}{k^3}\right) \right) \sim \frac{1}{6k^3} \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has positive terms, converges by limit comparison with the series of $1/(6k^3)$.

Summary

Passing from the discrete to the continuous:

- ▶ Asymptotic equivalences of functions also hold for sequences, taking as $\varepsilon(x)$ an infinitesimal sequence (for example $1/k$)
- ▶ Notable equivalences and Maclaurin expansions provide the asymptotic term

Strategy:

- ▶ Find a simple b_k with $a_k \sim b_k$, typically $b_k = c/k^\alpha$
- ▶ Conclude with the limit comparison test and the generalized harmonic series

Root test

Introduction: Root test

- ▶ The **root test** compares the series with a **geometric series** by studying the limit of $\sqrt[k]{a_k}$
- ▶ It is particularly useful when the general term contains **k -th powers**
- ▶ We will apply it to the series $\sum k^\beta b^k$ and $\sum b^k/k^k$

Theorem 5: Root test

Let $\{a_k\}$ be a sequence with non-negative terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell \in \mathbb{R}^*$$

then the series

$$\sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof (1/4)

Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell < 1$.

Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2}$$

Moreover, since $\ell < 1$, we have $\ell < 1 - \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \leq \ell + \frac{\varepsilon}{2} < (1 - \varepsilon) + \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2}$$

hence

$$a_k < \left(1 - \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

Proof (2/4)

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^k$ eventually.

The geometric series:

$$\sum_{k=0}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^k \quad \text{is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Proof (3/4)

Suppose that $\lim_{k \rightarrow +\infty} \sqrt[k]{a_k} = \ell > 1$.

Since $\sqrt[k]{a_k} \rightarrow \ell \in \mathbb{R}$, for every $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2}$$

Moreover, since $\ell > 1$, we have $\ell > 1 + \varepsilon$ for a suitable $\varepsilon > 0$. For this ε we therefore have, eventually:

$$\sqrt[k]{a_k} \geq \ell - \frac{\varepsilon}{2} > (1 + \varepsilon) - \frac{\varepsilon}{2} = 1 + \frac{\varepsilon}{2}$$

hence

$$a_k > \left(1 + \frac{\varepsilon}{2}\right)^k, \quad \forall k \geq m$$

Proof (4/4)

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^k$ eventually.

The geometric series:

$$\sum_{k=0}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^k \quad \text{is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

The cases $\ell = -\infty$ and $\ell = +\infty$

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

The series $\sum k^\beta b^k$

Observation 5

The series $\sum_{k=1}^{\infty} k^\beta \cdot b^k$ with $\beta \in \mathbb{R}$, $b \geq 0$ is

$$\left\{ \begin{array}{ll} \text{convergent} & \text{if } b < 1 \\ \text{divergent} & \text{if } b > 1 \\ \text{convergent} & \text{if } b = 1 \text{ and } \beta < -1 \\ \text{divergent} & \text{if } b = 1 \text{ and } \beta \geq -1 \end{array} \right.$$

Proof

It is a series with non-negative terms, and we have:

$$\sqrt[k]{k^\beta \cdot b^k} = b \cdot k^{\beta/k}$$

and

$$\lim_{k \rightarrow +\infty} k^{\beta/k} = \lim_{k \rightarrow +\infty} \exp \left(\underbrace{\frac{\beta}{k} \cdot \log k}_{\rightarrow 0} \right) = 1$$

hence, by the root test, it converges for $b < 1$ and diverges for $b > 1$. If $b = 1$, the series becomes:

$$\sum_{k=1}^{\infty} k^\beta = \sum_{k=1}^{\infty} \frac{1}{k^{-\beta}}$$

that is, a generalized harmonic series; it is convergent for $\beta < -1$, while it is divergent for $\beta \geq -1$. □

The series $\sum b^k/k^k$

Observation 6

The series $\sum_{k=1}^{\infty} b^k/k^k$ with $b \geq 0$ is convergent.

Proof. It is a series with non-negative terms, and we have:

$$\sqrt[k]{\frac{b^k}{k^k}} = \frac{b}{k} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

hence the series converges by the root test. □

Summary

Root test:

- ▶ If $a_k \geq 0$ and $\sqrt[k]{a_k} \rightarrow \ell$: the series converges if $\ell < 1$ and diverges to $+\infty$ if $\ell > 1$
- ▶ The proof compares a_k with a geometric series with ratio $1 \mp \frac{\varepsilon}{2}$

Applications:

- ▶ $\sum k^\beta b^k$: converges if $b < 1$, diverges if $b > 1$; if $b = 1$ it is a generalized harmonic series
- ▶ $\sum b^k / k^k$ converges for every $b \geq 0$

Ratio test

Introduction: Ratio test

- ▶ The **ratio test** studies the limit of the ratio of two consecutive terms a_{k+1}/a_k
- ▶ In this case too, the series is compared with a **geometric series**
- ▶ It is particularly useful when the general term contains **factorials**: we will use it for the series $\sum \frac{1}{k!}$

Ratio test

Theorem 6: Ratio test

Let $\{a_k\}$ be a sequence with positive terms. If the following limit exists:

$$\lim_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} = \ell \in \mathbb{R}^*$$

then the series

$$\sum a_k \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } \ell < 1 \text{ or } \ell = -\infty \\ \text{divergent to } +\infty & \text{if } \ell > 1 \text{ or } \ell = +\infty \end{cases}$$

Proof (1/4)

Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell < 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} < \left(1 - \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$\begin{aligned} a_{k+1} &< \left(1 - \frac{\varepsilon}{2}\right) a_k < \left(1 - \frac{\varepsilon}{2}\right) \left(1 - \frac{\varepsilon}{2}\right) a_{k-1} \\ &< \dots < \left(1 - \frac{\varepsilon}{2}\right)^{k-m+1} a_m \end{aligned}$$

Proof (2/4)

We have thus proved that $a_k < \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 - \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is convergent since } 1 - \frac{\varepsilon}{2} < 1$$

Hence, by the comparison test, the original series converges.

Proof (3/4)

Suppose that $\lim_{k \rightarrow +\infty} a_{k+1}/a_k = \ell > 1$. Reasoning as in the proof of the root test, there exists $m \in \mathbb{N}$ such that $\forall k \geq m$:

$$\frac{a_{k+1}}{a_k} > \left(1 + \frac{\varepsilon}{2}\right)$$

for a suitable $\varepsilon > 0$. Reasoning iteratively, this implies that:

$$\begin{aligned} a_{k+1} &> \left(1 + \frac{\varepsilon}{2}\right) a_k > \left(1 + \frac{\varepsilon}{2}\right) \left(1 + \frac{\varepsilon}{2}\right) a_{k-1} \\ &> \dots > \left(1 + \frac{\varepsilon}{2}\right)^{k-m+1} a_m \end{aligned}$$

Proof (4/4)

We have thus proved that $a_k > \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m$ eventually. The geometric series:

$$\sum_{k=m}^{\infty} \left(1 + \frac{\varepsilon}{2}\right)^{k-m} a_m \text{ is divergent since } 1 + \frac{\varepsilon}{2} > 1$$

Hence, by the comparison test, the original series diverges. □

The cases $\ell = -\infty$ and $\ell = +\infty$

Clearly, these arguments remain valid also if $\ell = -\infty$ or $+\infty$, respectively.

The series $\sum 1/k!$

Observation 7

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e$$

Proof. We have

$$\frac{1/(k+1)!}{1/k!} = \frac{1}{k+1} \rightarrow 0 \quad \text{as } k \rightarrow +\infty$$

Therefore the series, which has non-negative terms, converges by the ratio test. The proof that the sum of the series is equal to e will be given later using series of functions. □

Summary

Ratio test:

- ▶ If $a_k > 0$ and $a_{k+1}/a_k \rightarrow \ell$: the series converges if $\ell < 1$ and diverges to $+\infty$ if $\ell > 1$
- ▶ Iterating the estimate on the ratio, a_k is compared with a geometric series

Application:

- ▶ The series $\sum_{k=0}^{\infty} \frac{1}{k!}$ converges (and its sum is e)

Series with non-positive terms

Introduction: Series with non-positive terms

- ▶ The tests seen so far require non-negative terms
- ▶ We see that they easily extend to series whose terms **eventually** have the same sign

Extension of the tests

- ▶ We know that the behavior of a series does not change if we alter a finite number of its terms. Consequently, the tests for series with non-negative terms also apply to series whose terms are eventually non-negative.
- ▶ Moreover, by factoring a minus sign out of the whole series, we see that these tests can also be applied to series with non-positive terms, and hence to series whose terms are eventually non-positive.

In summary, therefore, the tests we have seen apply to series all of whose terms (except for a finite number) have the same sign.

Series with terms of constant sign:

- ▶ Altering a finite number of terms does not change the behavior of a series
- ▶ By factoring out a minus sign, the tests also hold for series with non-positive terms
- ▶ The tests apply to series all of whose terms, except for a finite number, have the same sign

Key messages

Summary (1/2)

1. Series with non-negative terms:

- ▶ They are regular: they converge if and only if the partial sums are bounded

2. Comparison tests:

- ▶ Comparison: $0 \leq a_k \leq b_k$ eventually, convergence passes from the dominating series and divergence from the dominated series
- ▶ Limit comparison: $a_k \sim b_k \Rightarrow$ same behavior (not the same sum)

3. Condensation test:

- ▶ For decreasing $\{a_k\}$: $\sum a_k$ and $\sum 2^k a_{2^k}$ have the same behavior

Summary (2/2)

4. Generalized harmonic series:

- ▶ $\sum \frac{1}{k^\alpha}$ diverges if $\alpha \leq 1$ and converges if $\alpha > 1$

5. Root and ratio tests:

- ▶ Limit ℓ of $\sqrt[k]{a_k}$ or of a_{k+1}/a_k : convergence if $\ell < 1$, divergence if $\ell > 1$

6. Extension:

- ▶ All the tests hold for series whose terms eventually have the same sign